



Hierarchical decision-making model for multi-aircraft formation in forest firefighting missions

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ABSTRACT

Large-scale forest fires necessitate the execution of firefighting missions by aircraft in formation. Enhanced collaboration both between and within formations can improve the efficiency of suppressing and extinguishing forest fires. Traditional research on fleet collaboration primarily employs optimization algorithms for target allocation, often overlooking the collaborative decision-making process among aircraft during water release operations. This paper addresses this gap by analyzing the constraints of the water release process and integrating firefighting tactics to propose a hierarchical decision-making model for multi-aircraft formations. The model introduces decision-making behaviors both between and within formations into the water release process, thereby strengthening the strategic alignment between water release mission planning and firefighting practices. Finally, based on the proposed decision-making model, simulations of forest firefighting scenarios involving multiple aircraft and formations are conducted to evaluate the impact of different formation structures on firefighting effectiveness.

1. Introduction

Forest fires are a major type of natural disaster that can have severe impacts on human production and daily life. Therefore, early intervention and rapid suppression of forest fires have always been a focal point of concern for countries worldwide. Aircraft, being less constrained by terrain, can quickly deliver fire retardants to fire-affected areas to curb the spread of flames, playing a crucial role in forest firefighting missions [1]. With the increasing deployment of fixed-wing aircraft and helicopters, aerial forest firefighting has evolved from single-aircraft operations to multi-aircraft and multi-formation approaches. This evolution has significantly increased the complexity of missions, demanding higher levels of coordination both between formations and among individual aircraft.

Research on aerial forest firefighting includes inductive summarization based on historical cases and data [1–5], as well as mission simulation-based deduction [6–9]. The primary focus of studies based on historical cases and data is to deploy aircraft at different airports before the occurrence of forest fires, thereby ensuring the fastest aerial firefighting response within a certain period for the areas covered by these aircraft [10–12]. Alternatively, such studies evaluate and select aircraft for aerial firefighting missions [13,14]. Research based on

simulation methods, on the other hand, can simulate tasks such as aircraft monitoring fire conditions and dispensing fire retardants [15–18], based on forest fire spread models, to design and optimize aircraft firefighting missions [6,8,19,20]. However, research centered on aircraft firefighting missions primarily focuses on issues such as the coordination and task allocation of reconnaissance drone swarms. For manned aircraft, such as fixed-wing planes and helicopters, the main research areas include flight safety and water release methods [21–23]. Only a small portion of studies address multi-aircraft formation firefighting missions in forest fire scenarios [9,24,25]. Among these, Prakash et al. categorized multi-aircraft forest firefighting missions as system-of-systems tasks, emphasizing the need for coordination between aircraft and between aircraft and ground firefighting equipment [24]. The distinction between collaborative firefighting missions involving multi-aircraft formations and single-aircraft missions lies not only in the number of aircraft but also in the emergent characteristics of the formation. Based on Maier's definition of systems [26], a group of aircraft exhibits system-of-systems characteristics, possessing unique attributes that individual aircraft do not. While there is considerable research on single-aircraft firefighting missions, studies on multi-aircraft formation firefighting are relatively scarce and lack detail.

On the other hand, strategies for aerial forest firefighting have been

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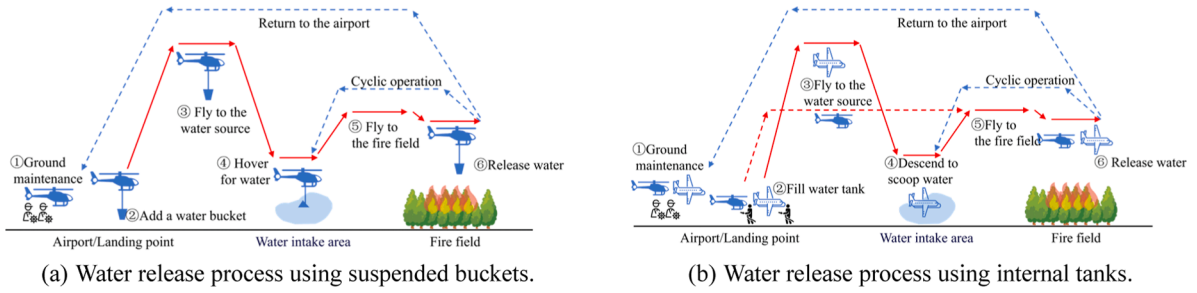


Fig. 1. Water release processes: suspended buckets and internal tanks.

summarized in relevant technical specifications [27], operational manuals [28], and other guidelines. Zhang et al. summarized helicopter formation bucket water-dropping operations, clarifying the command relationship between the lead aircraft and wingmen in a formation. Based on practical aerial forest firefighting experiences, they proposed methods and tactics for formation firefighting, representing a further development of single-aircraft firefighting tactics [29]. In both real-world aerial firefighting operations and human-in-the-loop simulation exercises, water release decisions for aircraft can be guided by commanders to achieve specific tactical objectives. However, in human-out-of-the-loop simulation studies of formation firefighting missions, modeling and simulation at the formation level often overlook the integration of firefighting strategies and inter-formation collaboration.

In the research on multi-aircraft formation forest firefighting missions, previous studies still have the following areas for improvement:

1. Selection of suitable aircraft for firefighting missions: The selection process relies heavily on historical cases and data, which are predominantly based on single-aircraft operations. This approach overlooks the emergent properties that arise in multi-aircraft formations.
2. Lack of focus on inter-aircraft collaboration: The methods and tactics for formation firefighting are not adequately reflected in mission simulations, leading to discrepancies between simulation results and actual firefighting practices.
3. Quantitative decision-making basis for formation firefighting tactics: There is a lack of quantifiable decision-making criteria, making it difficult to apply these tactics to multi-round simulation exercises or firefighting scenarios under different conditions.

This study addresses the shortcomings in existing research on multi-aircraft formation forest firefighting missions by proposing a two-layer decision-making method for multi-aircraft formations. The method integrates technical specifications and firefighting practical experience to achieve collaborative tactical decision-making between and within

formations. Firefighting process constraints are incorporated to define quantitative evaluation metrics and calculation methods for assessing tactical choices, while a quadtree-based approach is applied for fire scene data discretization. Optimization algorithms are utilized for weight assignment, and the proposed decision-making method is tested for firefighting effectiveness in multi-aircraft formations. Aircraft formation construction strategies are further analyzed based on simulation results. The primary contributions of this study are summarized as follows:

1. Definition of decision metrics: Decision metrics adhering to firefighting safety constraints and mission objectives are established. Calculation methods and fire scene discretization processes are proposed to quantify the tactical decision-making process.
2. Formulation of a formation-level decision model: A firefighting decision model for formations is developed by assigning weights to evaluation metrics through optimization algorithms.
3. Enhanced tactical collaboration in simulations: The decision model is applied to multi-aircraft forest firefighting simulations, demonstrating stronger tactical collaboration between and within formations compared to prior approaches. Multi-round simulations test the firefighting effectiveness of formation structures, replacing manual experience-based guidance with quantitative data-driven insights.

This paper is organized as follows: Section 2 describes the research problem and presents the methodological framework. Section 3 analyzes and models single-aircraft strategies and inter-aircraft strategies. Section 4 details simulation experiments and analysis of results for the proposed method. Section 5 concludes the paper.

2. Problem statement and method structure

Various types of aircraft are capable of performing aerial firefighting missions. Among them, Very Large Air Tankers (VLATs), Large Air Tankers (LATs), Heavy Heli Tankers (HHTs), and Amphibious Water Scooper Aircraft (AWSA) are often tasked with dispensing fire

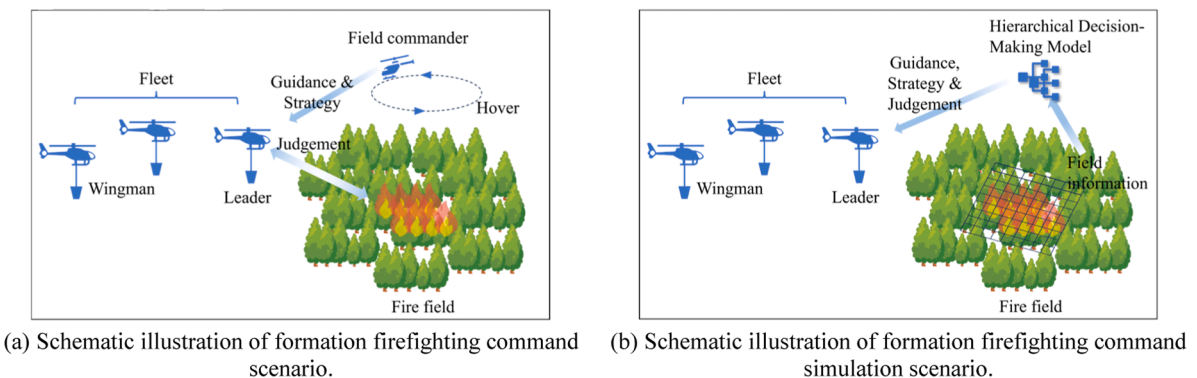


Fig. 2. Schematic illustration of water release strategy execution.

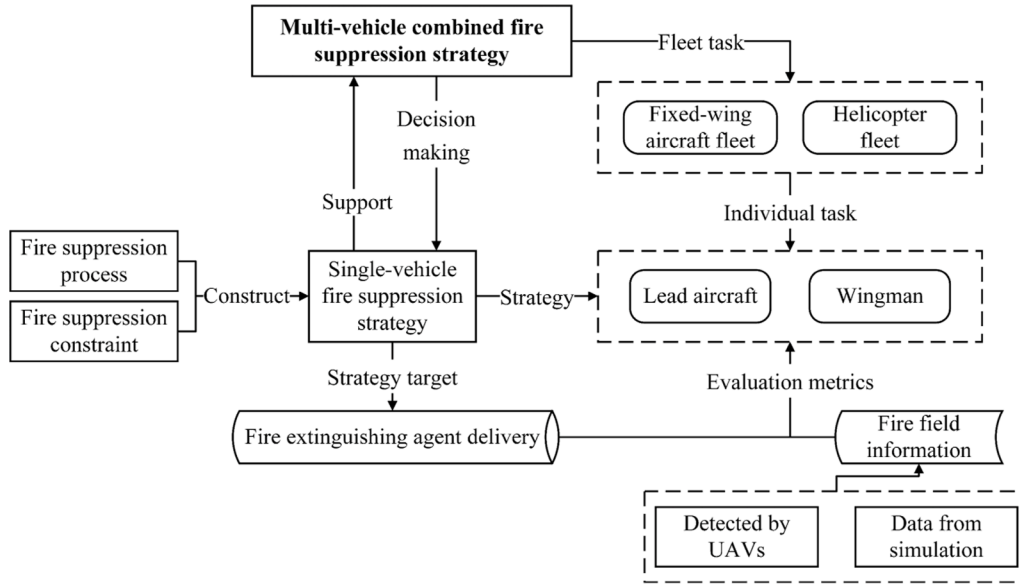


Fig. 3. Two-layer firefighting strategy framework for multi-aircraft forest firefighting systems.

retardants, which directly impact the fire scene. Fire retardants are primarily transported and released through externally suspended buckets or internal tanks carried by the aircraft. The complete mission process is illustrated in Fig. 1. Among these tasks, the fire retardant release process is the most critical operation influencing fire spread, and this paper focuses on decision-making for this process. The remaining tasks are treated as boundary constraints for subsequent simulations.

Other aircraft involved in aerial firefighting include light helicopters and unmanned aerial vehicles (UAVs). These aircraft perform tasks such as fire scene reconnaissance and fire retardant effectiveness assessment by hovering over the fire-affected area for extended periods. In this paper, these aircraft are considered sources of fire scene information. Under the assumption that the fire scene status is fully monitored and information transmission is unaffected by terrain, decisions regarding the water release process are made.

Another important component of forest firefighting missions is ground firefighting forces, including firefighters and firefighting vehicles. These forces access the fire-affected area via roads or through helicopter landings, rappelling, or skid descents. When aircraft execute water release tasks, the presence of ground personnel must also be considered. Water may need to be dropped in their vicinity to provide support or rescue. In this paper, ground firefighting forces are simplified, with only their relative positions to the fire scene considered to determine whether support is required.

The water release process for multi-aircraft formations aligns with the aforementioned workflow. In practical firefighting operations, a light helicopter typically hovers over the fire-affected area to serve as the on-site command unit, as illustrated in Fig. 2.(a). This aircraft integrates real-time fire scene data to communicate guidance information for the firefighting zone and designated tactical strategies to the formation. The lead aircraft and wingmen then execute water release operations according to predefined tactics, while specific water release points are autonomously determined by individual aircraft. In human-out-of-the-loop simulation environments, the formation strategy is replaced by the two-layer decision-making model proposed in this study. By analyzing current fire scene information, the model generates formation-level firefighting strategies and provides water release location judgments for individual aircraft, as depicted in Fig. 2.(b).

For forest fire spread, this paper adopts a discretization approach in the horizontal two-dimensional plane, dividing the area into equal-sized square regions. These regions are categorized into forest areas (unburned regions), burning areas, and burned areas. This discretization

method transforms the fire spread area into an analyzable and evaluable object while maintaining consistency with forest fire spread simulation models. As a result, the use of fire spread simulations to validate the proposed method in this paper is both feasible and reliable. However, since the forest fire spread model is not the primary focus of the paper, the cellular automata-based forest fire spread model developed by Liu et al. is directly adopted [18]. On the other hand, extensive prior research has focused on aircraft deployment strategies to determine the types and quantities of aircraft involved in firefighting missions. Furthermore, during large-scale fire incidents, reinforcements are typically dispatched from other regions using additional aircraft. Therefore, this study treats the formation structure as a predefined condition and investigates aircraft mission decision-making after firefighting operations commence. When only one formation is deployed, certain formation-level strategies are excluded from consideration.

This paper addresses the water release process in multi-aircraft formation forest firefighting missions by proposing a two-layer decision-making model that spans from the formation level to the individual aircraft level. The model enables task decision-making within and between formations, taking into account the presence of other formations and aircraft. This approach enhances the alignment between simulation experiments and actual firefighting strategies. The proposed multi-aircraft formation decision-making model is illustrated in Fig. 3.

The single-aircraft strategy is constructed based on operational workflows and mission constraints, serving as the fundamental building block of the multi-aircraft strategy. It defines the specific tactical methods and tasks for individual aircraft.

The multi-aircraft strategy evaluates and makes decisions based on the single-aircraft strategies. At the formation level, it determines the overall formation tasks and assigns specific tasks to individual aircraft based on their single-aircraft strategies. This process ultimately determines the decision outcomes, including water release locations.

The decision outcomes of each aircraft, along with fire scene information, serve as input conditions for task execution, enabling collaborative effects within the formation. This iterative process ensures that the actions of individual aircraft are coordinated and aligned with the overall mission objectives. By integrating single-aircraft and multi-aircraft strategies, the proposed model enhances the consistency between simulation experiments and real-world firefighting practices, providing a robust framework for optimizing aerial firefighting operations. The weights of decision metrics in this study are derived through optimization algorithms, though manual weight assignment remains a

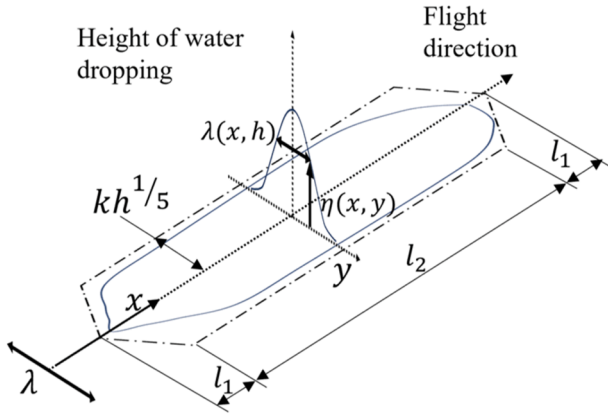


Fig. 4. Schematic of water dispersion pattern.

feasible alternative. In practical applications, this decision model can serve as a reference for decision-making in human-in-the-loop simulations. Conversely, human decisions can also be integrated as inputs to the decision model.

3. Model

The modeling of the strategy is based on the constraints of the water release process, including flight state constraints, water distribution constraints, and decision-making process constraints. On this foundation, the decision-making results for single-aircraft water release are obtained by evaluating the fire scene status. Furthermore, based on multi-aircraft collaborative firefighting tactics and the current role of the aircraft within the formation, the water release tactics and locations are determined.

3.1. Constraints of the water release process

The water release process over the fire scene consists of three main phases: the arrival phase, the water release phase, and the departure phase. Key operational constraints include aircraft speed limitations, water discharge constraints, and climb angle constraints during departure. The speed constraints primarily affect the aircraft's time allocation during phases such as cruising, which is critical for mission simulation duration calculations. The water discharge constraints serve dual purposes: first, to calculate the distribution of discharged water post-release, and second, to influence the horizontal flight distance over the fire scene. Climb angle constraints, acting as flight safety limitations,

are applied to eliminate water release locations that violate safe operational requirements.

For helicopters equipped with external buckets, the bucket is connected to the aircraft via a suspension line. This configuration imposes higher demands on the helicopter's flight maneuverability. To ensure flight safety, specific operational requirements must be met during this phase as follows:

$$v_{\max} = \begin{cases} 140\text{km/h, without water} \\ 160\text{km/h, carrying water} \\ 80\text{km/h, drop water} \end{cases} \quad (1)$$

$$\phi_{\max} = \begin{cases} 10^\circ, \text{ without water} \\ 15^\circ, \text{ carrying water} \end{cases} \quad (2)$$

where v represents helicopter's air speed, ϕ represents the helicopter's bank angle during a turn, and

$$R = \frac{|\mathbf{v}|^2}{g \cdot \tan \phi} \quad (3)$$

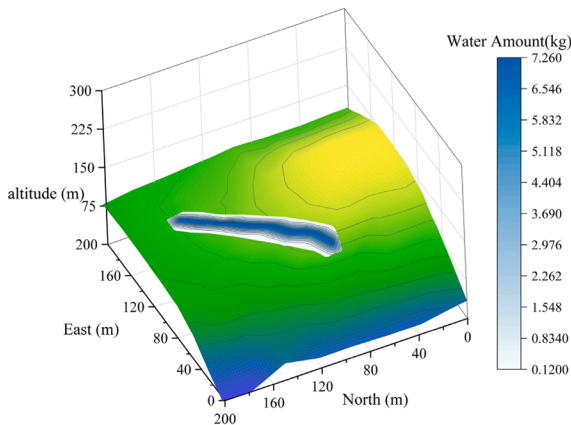
where R represents the turning radius.

During the departure phase, it is essential to ensure that the aircraft's climb gradient is greater than the local terrain slope to avoid collisions with the ground. Specifically, the climb angle γ and the aircraft's true altitude TA must satisfy the following conditions:

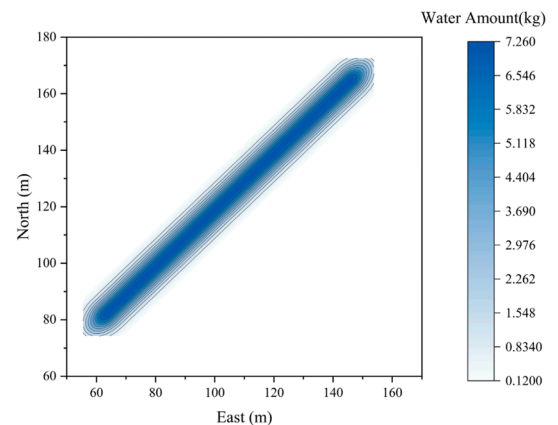
$$\begin{cases} \gamma \geq \text{slope}_{\text{terrain}} \\ TA \geq h_{\min} \end{cases} \quad (4)$$

The shape of the water dispersion pattern is referenced from previous studies and abstracted as shown in Fig. 4 [30]. When calculating the variation of the water release width along the flight direction, it is further simplified into a hexagonal shape, as indicated by the dashed line in the figure. The distribution of the water dispersion pattern in the width direction approximates a normal distribution. The width is related to the aircraft's airspeed and the descent velocity of the fire retardant. At the same flight speed, the maximum width λ of the water dispersion pattern is considered to be proportional to the fifth root of the water release height h . Based on the water dispersion pattern and the distribution of water volume, the along-flight direction distribution by Eq. (5) and the vertical distribution by Eq. (6) are constructed for the water release process.

$$\lambda(x, h) = \begin{cases} kh^{1/5} \cdot x/l_1, & x < l_1 \\ kh^{1/5}, & l_1 \leq x \leq l_1 + l_2 \\ kh^{1/5} \cdot (2l_1 + l_2 - x)/l_1, & l_1 + l_2 < x \leq 2l_1 + l_2 \end{cases} \quad (5)$$



(a) Spatial distribution of the water dispersion pattern.



(b) Planar distribution of the water dispersion pattern.

Fig. 5. Calculation results for water dispersion distribution.

$$\eta(x, y) = \frac{\lambda}{\sqrt{2\pi}\sigma^2} \exp\left(-\frac{y^2}{2\sigma^2}\right) \quad (6)$$

where k serves as the scaling coefficient to determine the maximum width of the water discharge zone at the current release altitude. Parameters l_1 and l_2 characterize the longitudinal distribution of water discharge along the release direction: l_1 represents the length of the segment with increasing or decreasing water volume, while l_2 denotes the length of the segment with uniform water volume. The water distribution at a distance y from the central axis of the discharge zone is modeled as $\eta(x, y)$, where σ is the standard deviation of the normal distribution, satisfying $\sigma = \lambda/6$.

To prevent poor water dispersion due to excessive spreading, the water release height of the aircraft must be constrained. Previous studies have summarized the water release heights for different types of aircraft [30], indicating that fixed-wing aircraft and helicopters typically operate at heights between 10 m and 80 m. In this paper, the water release height is set to 50 m above ground level to achieve the corresponding water dispersion width.

The water volume distribution is discretized to calculate the water volume per square meter. Fig. 5 illustrates the distribution of the water dispersion pattern over terrain and its planar spread when the water release length is 120 m, the designated width is 20 m, and the water is released in a northeast direction. The total water release volume is 4000 kg.

3.2. Single-aircraft strategy

The construction of the water release process and the selection of water release locations are based on the following steps:

- (1) Identify possible water release locations: Use the fire's burning positions as constraints to determine all potential water release locations.
- (2) Evaluate water release locations: Assess each location based on evaluation metrics, either obtaining evaluation results or excluding the point.
- (3) Select optimal water release point: Choose the location with the best evaluation result under the current strategy as the water release point and determine the flight path.

To identify possible water release locations, the fire-affected area is first enveloped. Due to factors such as terrain, vegetation distribution, and wind direction, fire spread often forms irregular shapes, which can be abstracted as fire lines and fire points. A burning area can be composed of such basic elements. For aircraft, the environment above the fire-affected area is influenced by high temperatures, leading to more intense convection and the presence of smoke, which can affect flight safety. Therefore, when processing the fire-affected area, the following principles are applied:

- (1) Flight over burning areas: Aircraft are only allowed to fly over burning areas briefly during water release and must avoid crossing these areas at other times.
- (2) Level flight for water release: Aircraft perform water release in level flight without executing maneuvers.

When evaluating water release points, the shape of the fire-affected area is first processed. For complex and irregular burning areas, an outer envelope convex polygon is calculated, and the edges of this polygon represent the set of possible water release points. When using drones or other methods for fire reconnaissance, the fire spread is described in terms of fire points and fire lines, which include information about the burning area's location. After abstraction and discretization, discrete fire points corresponding to the resolution are obtained, and the

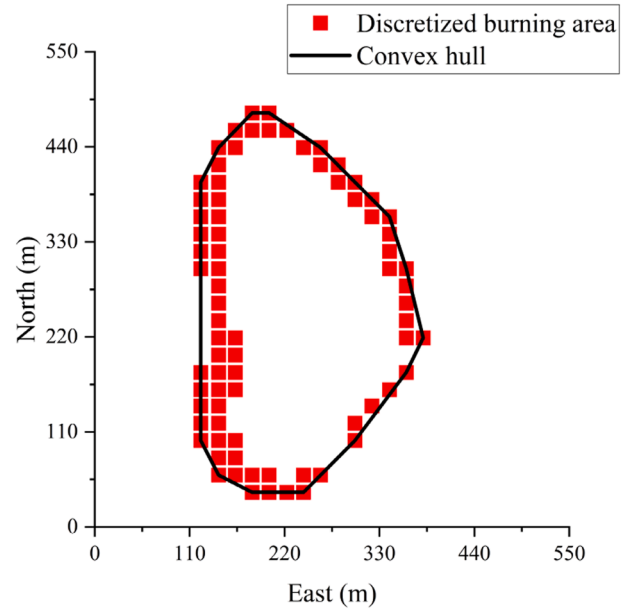


Fig. 6. Result of convex hull for burning areas.

envelope results are as shown in Fig. 6.

The discretization of burning areas is achieved through a grid-based partitioning approach, transforming continuous fire scenes into spatially segmented units for partitioned analysis, thereby reducing computational complexity. In practical applications, the resolution of fire scene discretization can reach meter-level precision, depending on the capabilities of monitoring equipment. However, higher-resolution discretization entails significantly increased computational demands. The resolution adopted in this study aligns with the forest fire spread simulation model, as finer-than-resolution details cannot be captured by the simulation framework.

By solving the convex hull of the burning area's outer envelope, the potential flight region for aircraft water release is narrowed down. In actual forest fire scenarios, multiple ignition points and burning areas often exist. Therefore, for a forest fire area FA with $n(n \geq 1, n \in N)$ distinct burning regions, the area can be represented using a set of discretized fire points, as follows:

$$FA = \{fp_1^1, fp_2^1, \dots, fp_i^1, \dots, fp_1^n, fp_2^n, \dots, fp_j^n\} \quad (7)$$

where $fp = (x_f, y_f)$ represents the positions of the discretized fire points, with the superscript indicating the burning region number and the subscript indicating the point's index within that region.

After calculating the convex hull of the burning regions, FA can be simplified into polygonal regions with number of n for subsequent water release process calculations. The convex hull set $CH^k(fp_p^k, fp_q^k, \dots, fp_r^k)$, $fp_p^k, fp_q^k, \dots, fp_r^k \in FA$ represents the closed polygonal region formed by sequentially connecting the points $fp_p^k, fp_q^k, \dots, fp_r^k$, satisfying:

$$\forall fp_i^k \in FA, fp_i^k \odot CH^k(fp_p^k, fp_q^k, \dots, fp_r^k) \quad (8)$$

where $\odot$ indicates that a point lies inside or on the boundary of the closed polygonal region $CH^k(fp_p^k, fp_q^k, \dots, fp_r^k)$, and the superscript k denotes the convex hull set corresponding to the burning region. The set of feasible water release regions WA can then be expressed as:

$$\forall CH^i(fp), WA \cap CH^i(fp) = \emptyset \quad (9)$$

To account for the influence of wind on the water release landing points during the process, the convex hull node positions need to be

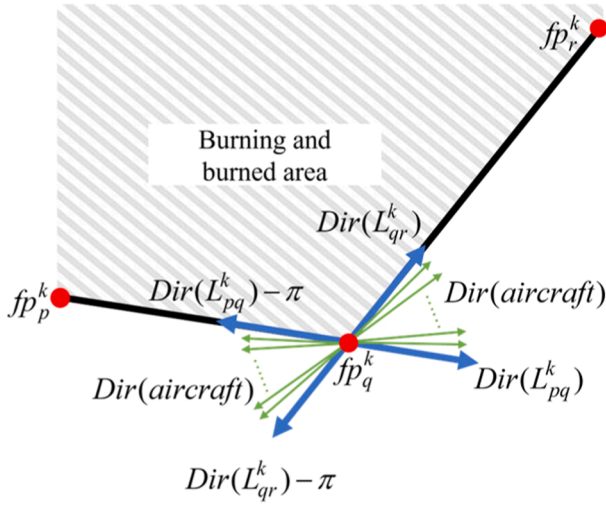


Fig. 7. Range of water release directions at endpoints.

adjusted based on wind direction. This ensures that the aircraft's water release position is located in the upwind area of the fire. Defining the boundary set $CHB^k(L_{pq}^k, \dots, L_{rp}^k)$ of $CH^k(fp_p^k, fp_q^k, \dots, fp_r^k)$, where L_{pq}^k represents the line segment with endpoints fp_p^k and fp_q^k , and using $Dir(x)$ to denote the direction, the offset is calculated as:

$$\lambda_{wind} \cdot \text{Dot}(Dir(L_{pq}^k), Dir(wind)) + \mu_{wind}, L_{pq}^k \in CHB^k \quad (10)$$

where λ_{wind} is the offset correction coefficient, and μ_{wind} is the base offset.

In aerial firefighting practice, to achieve direct fire suppression, control fire intensity, and slow down the spread, the direction of water release by aircraft is generally parallel to the fire line. This approach aims to maximize coverage of the fire line or the area in the direction of fire spread. When determining the water release position and direction for the aircraft, two scenarios are discussed:

(1) Fixed Direction with Undetermined Release Position: In this case, the aircraft flies along a boundary segment of the convex hull and begins water release at a selected point on the segment. The flight direction during water release must align with or oppose the direction of the boundary segment at the target position, as follows:

$$Dir(aircraft) = Dir(L_{pq}^k) \text{ or } Dir(L_{qr}^k) - \pi, L_{pq}^k \in CHB^k \quad (11)$$

(2) Fixed Release Position with Undetermined Direction: In this case, the aircraft performs water release centered on a specific point within $fp_p^k, fp_q^k, \dots, fp_r^k$. Since the boundary directions on either side of the endpoint differ, the aircraft must avoid flying over the fire-affected area. Thus, the flight direction must lie between the directions of the two adjacent boundary segments, as illustrated in Fig. 7.

These considerations ensure that the water release process is optimized for effective fire suppression while maintaining flight safety.

When the aircraft reaches different positions or arrives at the same position from different directions, the trajectory length varies. Assuming a constant flight speed, the arrival time of the aircraft corresponds to its position and direction. This means that at any given moment, the feasible water release positions for the aircraft are uniquely determined along the boundaries or endpoints. Thus, a time-dependent set of feasible water release positions can be obtained.

Define the potential water-dropping point set at the current moment as PWP_t , where each element in the set consists of a water release position $wp = (x_w, y_w)$ and a water release direction Dir . Each element must satisfy the constraints on position and direction mentioned earlier. Therefore, we have:

$$PWP_t = \{(wp_1, Dir_1), (wp_2, Dir_2) \dots (wp_n, Dir_n)\} \quad (12)$$

Therefore, at any given moment, it is only necessary to evaluate the points in PWP_t . Based on the criteria of fire spread containment effectiveness, flight safety, and ground target safety, the following evaluation metrics for water release positions are proposed, as illustrated in Fig. 8.

The heat transfer process of forest fire spread is influenced by wind and terrain [18]. To ensure that water release covers a longer fire line or creates a larger barrier area in the direction of fire spread, the following conditions are proposed:

$$e1 = \overrightarrow{Dir_{aircraft}} \cdot \overrightarrow{Dir_{wind}} \quad (13)$$

$$e2 = \overrightarrow{Dir_{aircraft}} \cdot \frac{(\partial h / \partial x, \partial h / \partial y)}{\|(\partial h / \partial x, \partial h / \partial y)\|} \quad (14)$$

where $(\partial h / \partial x, \partial h / \partial y)$ represents the local gradient vector of the water release area. For $e1$ and $e2$, the water release achieves the best effect when the directional angle is 90° .

The metric $e3$ addresses the potential "flying fire" phenomenon,

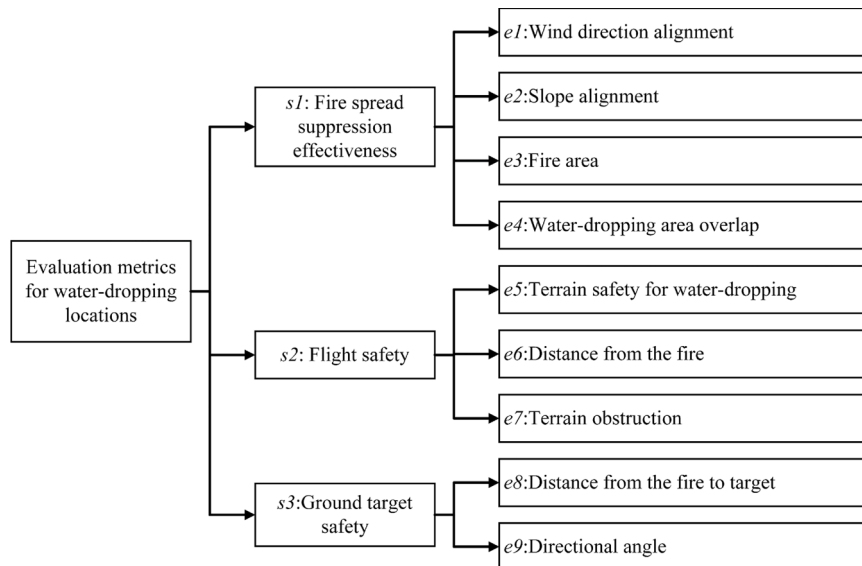


Fig. 8. Evaluation metrics for single-aircraft water release position selection.



Fig. 9. Concentrated water release pattern.

where burning debris or high-temperature embers are carried into the air by flame plumes or convective columns and driven by environmental winds to ignite new burning areas in unburned regions. For new burning areas that may arise during fire spread, the response logic aligns with that of forest fires, emphasizing the initial suppression time. Thus, the size of the burning area indirectly represents the duration of burning in different regions.

The metric $e4$ characterizes the overlap between the current water release position and the previous water release positions of other aircraft in the formation. It is represented by the sum of the overlapping areas between the current and past water release regions:

$$e4 = \sum S_{\text{overlap}} / S_{\text{water}} \quad (15)$$

Flight safety includes both ground and aerial impacts. The metric $e5$ evaluates the influence of terrain on flight safety during the water release phase, using the variance of terrain height along the projected flight path as an indicator:

$$e5 = \frac{1}{n} \sum_{i=1}^n (h_i - \bar{h})^2 \quad (16)$$

The metric $e6$ represents the minimum distance between the aircraft's flight path projection on the ground and the fire-affected area, measuring the extent to which the aircraft is affected by the fire and its flow field during flight. The metric $e7$ reflects Eq. (4); if terrain affects the aircraft's ability to depart from the fire scene, the point cannot be used as a water release area.

Ground targets $gt = (x_g, y_g)$ are valuable facilities that require protection. The metric $e8$ is the minimum distance between the current water release position and all ground targets:

$$e8 = \min(\text{Dis}(wp, gt_i)), gt_i \in GT \quad (17)$$

where GT is the set of all ground targets requiring protection.

For the fire-affected area, the closest distance to the ground targets is denoted as $\text{Dis}(fp_{\min D}, gt_j)$, and the corresponding point is $fp_{\min D}$:

$$\forall fp_i \in FA, gt_j \in GT, \text{Dis}(fp_{\min D}, gt_j) \leq \text{Dis}(fp_i, gt_j) \quad (18)$$

Thus, for $e9$, we have:

$$e9 = \frac{fp_{\min D} - wp}{\|fp_{\min D} - wp\|} \cdot \frac{fp_{\min D} - gt}{\|fp_{\min D} - gt\|} \quad (19)$$

When the result is positive, it indicates that the water release position lies in the direction of fire spread toward the ground target, and the water release can block this spread trend.

When evaluating each element in PWP_t , the evaluation result r_i for each element is:

$$r_i = W \times E_i^T \quad (20)$$

where $W = [w1, w2, \dots, w9]$ is the weight of the evaluation metric, and $E_i = [e1, e2, \dots, e9]$ is the evaluation result for the current element.

For calculating distance metrics in large-scale fire scenarios, the number of discretized burning positions fp is substantial, and most regions do not significantly affect the results. A quad-tree indexing method is employed for more efficient searching. Quad-tree data partitioning and indexing methods are widely used, and this paper adopts their basic

approach without detailed elaboration [31,32].

3.3. Formation water release strategy

The multi-aircraft collaborative firefighting strategy builds on the single-aircraft firefighting tactics and evaluation metrics, focusing on coordination and scheduling between aircraft, formations, and ground personnel. This strategy includes multi-aircraft water release strategies and air-ground collaboration strategies. The multi-aircraft water release strategy determines water release positions by considering coordination between formations, individual aircraft, and collaboration between aircraft and ground personnel. It encompasses three main approaches: concentrated water release, dispersed water release, and fixed-wing-helicopter collaborative water release.

3.3.1. Concentrated water release

This strategy applies to both fixed-wing aircraft formations and helicopter formations. It includes three sub-strategies: sequential water release, overlapping water release, and firebreak spraying.

- (1) Sequential water release: Aircraft release water in a sequential manner to ensure continuous coverage of the fire line, as shown in Fig. 9.(a).
- (2) Overlapping water release: Aircraft release water in overlapping patterns to increase water density in critical areas, as shown in Fig. 9.(b).
- (3) Firebreak spraying: Similar to sequential water release, this strategy involves creating firebreaks by continuous water release. It is particularly useful for fixed-wing aircraft formations when dealing with complex fire lines or fires in challenging terrains like canyons, cliffs, or steep slopes.

When using concentrated water release strategy, the overlap metric in the single-aircraft evaluation contributes positively, encouraging subsequent aircraft to choose positions closer to previous release points.

3.3.2. Dispersed water release

This strategy is primarily used by helicopter formations in coordination with concentrated water release. The first echelon of helicopters disperses water across the fire-affected area to reduce temperatures and control fire spread, while subsequent echelons perform concentrated water release. In this strategy, the overlap metric contributes negatively, encouraging subsequent aircraft to choose positions farther from previous release points.

3.3.3. Fixed-wing and helicopter collaborative water release

In this strategy, fixed-wing aircraft formations perform concentrated water release on key areas of the fire, followed by helicopter formations providing supplementary water release to achieve targeted fire suppression. The overlap metric contributes positively, and the lead helicopter must consider this metric to ensure complementary water release with the preceding fixed-wing formation.

3.3.4. Air-ground collaboration

When ground personnel safety is threatened by the fire, the nearest

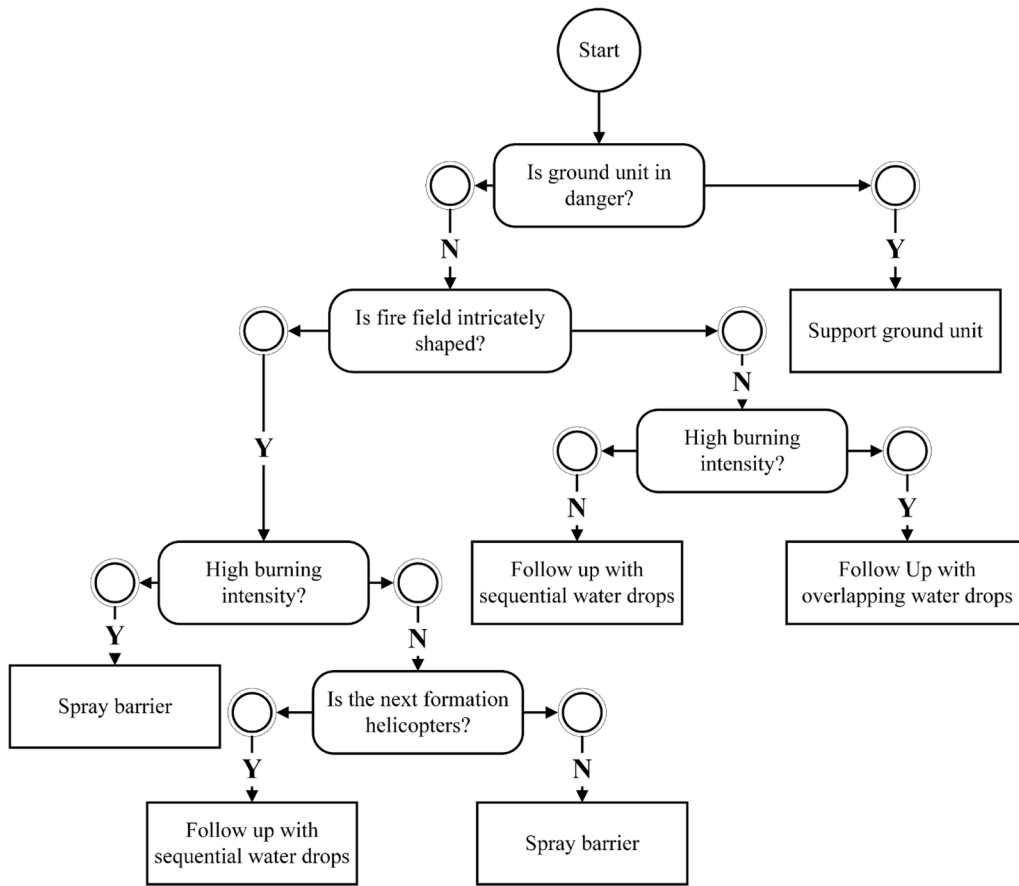


Fig. 10. Decision tree for fixed-wing aircraft formation water release.

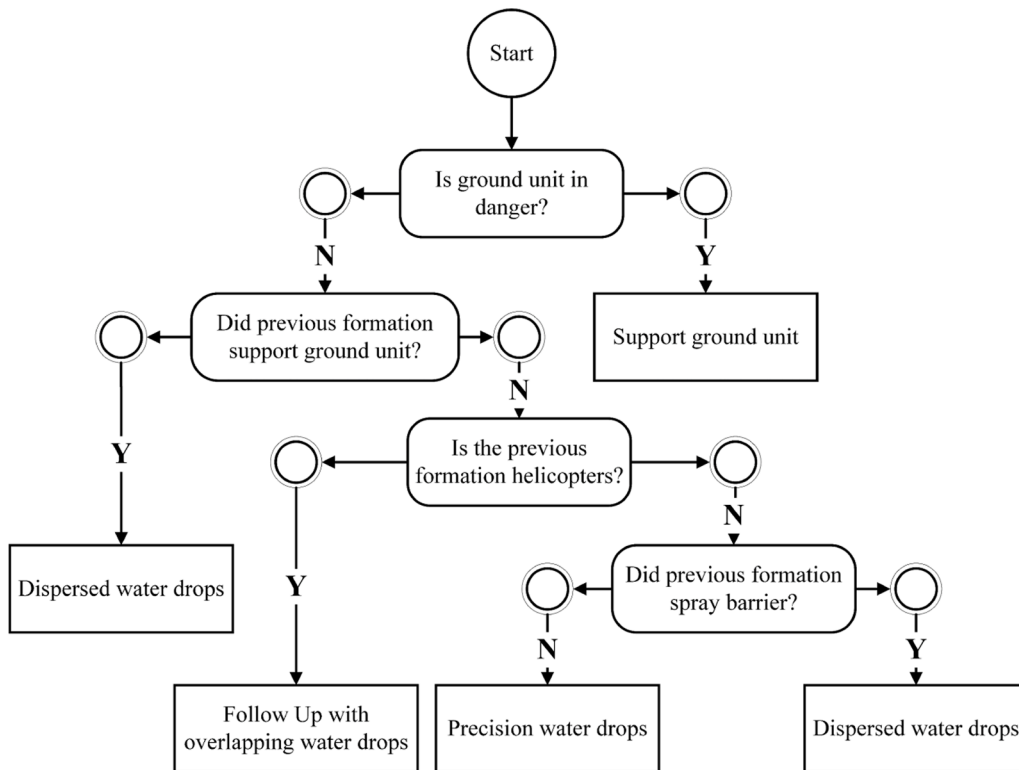


Fig. 11. Decision tree for helicopter formation water release.

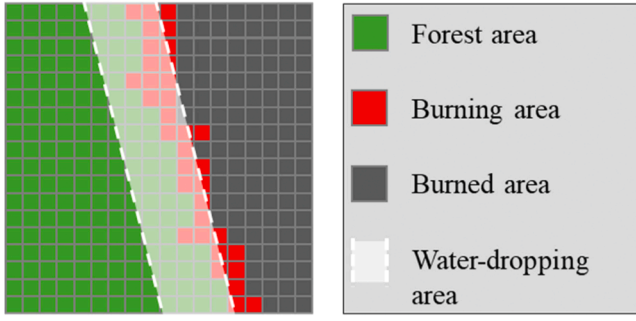


Fig. 12. Schematic of fire line complexity assessment.

aircraft or formation prioritizes their location for water release, executing either single-aircraft or concentrated water release strategies to ensure their safety. Ground personnel safety is the primary consideration in water release decision-making, ensuring rapid air-ground collaboration. Helicopter formations must also consider the water release strategies of preceding fixed-wing formations.

The need for ground personnel support or rescue is determined based on fire spread, personnel location, and the completion status of firebreaks. If the fire spreads within a critical distance of ground personnel and firebreaks are not yet established, air-ground collaborative water release tasks are initiated. Ground personnel are grouped, and each group is assumed to be at a single location for computational purposes.

The strategy is modeled using a decision tree approach, which is efficient and highly interpretable. This method determines the current formation's water release strategy and influences single-aircraft strategies. The decision trees for fixed-wing aircraft and helicopter formations are illustrated in Figs. 10 and 11, respectively. These single-attribute decision trees enable rapid decision-making for formation water release strategies. Attributes such as fire line complexity and fire intensity are evaluated and scored to guide decisions.

In Figs. 10 and 11, the evaluation of the fire scene includes two parts: fire line complexity assessment and fire intensity assessment. The fire line complexity is evaluated based on the projected distribution of the burning area, while the fire intensity is assessed using the proportion of burning area per unit area. These metrics are calculated as follows:

The fire line complexity FSC is determined based on the envelope region calculated using Eq. (8), which provides potential water release positions. Using the water release direction from Eq. (11) and the water dispersion width of the current aircraft, the possible coverage area of the water dispersion is determined. The complexity of the fire line is then assessed by analyzing the distribution of discretized burning regions within the water dispersion area, as illustrated in Fig. 12.

The complexity of the fire line is evaluated by computing the

reciprocal sum of the distances between the burning regions and the centerline of the water dispersion area. For the fire line distribution on L_{pq}^k , the following formula is used:

$$FSC^k = \sum_{i=1}^n 1/d_i \quad (21)$$

where d_i represents the distance between the i -th discretized burning region and the centerline of the water dispersion.

For discretized burning regions, a square with side length a is used for discretization. Eq. (21) can be normalized based on the maximum width λ of the water dispersion from Eq. (5) and the total possible water dispersion length L_{pq}^k along the envelope line:

$$FSC_0^k = \frac{L_{pq}^k}{a} \sum_{i=1}^n \frac{1}{a(i-0.5)} \frac{1}{n}, n = \text{floor}\left(\frac{\lambda - a/2}{a}\right) + 1 \quad (22)$$

$$\widehat{FSC}^k = FSC^k / FSC_0^k \quad (23)$$

The burning strength BS is calculated using a quad-tree method:

$$BS = \frac{nS_{unit}}{\sum_{i=1}^m S_i} \quad (24)$$

where $S_{unit} = a \cdot a$ is the area of the discretized burning region, n is the total number of burning regions, S_i is the area of quad-tree nodes containing burning regions, and m is the total number of such nodes.

To avoid additional node partitioning caused by outliers, which could lead to underestimation in Eq. (24), the quadtree partitioning method is modified to prioritize maximum depth, as shown in Fig. 13.

For the multi-aircraft water release decision model, the multi-aircraft collaboration and single-aircraft execution strategies involve multiple evaluation metrics. These metrics are constructed based on decision-making considerations from firefighting practices but lack structured data to support weight assignment. Commonly used methods, such as the Analytic Hierarchy Process (AHP), assign weights based on subjective factors like expert priorities and empirical knowledge [33,34]. To obtain evaluation weights suitable for execution strategies, a genetic algorithm is employed to optimize the weights of the evaluation metrics.

The genetic algorithm is an optimization technique based on the principles of natural selection and genetics. It simulates the process of biological evolution through operations such as selection, crossover, and mutation, gradually improving the quality of solutions. After each iteration, the algorithm retains partial information from the previous generation, preserving effective combinations of evaluation metric weights. Due to the sequential nature of fire spread simulations over time, evaluating the performance of a single weight set may require several minutes. Consequently, a trade-off must be made between the depth of weight space exploration and total computational time.

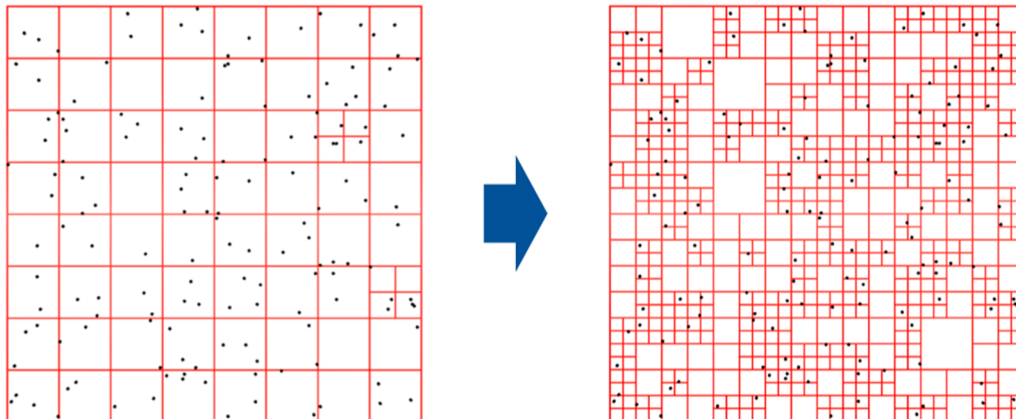


Fig. 13. Comparison of quad-tree partitioning methods.

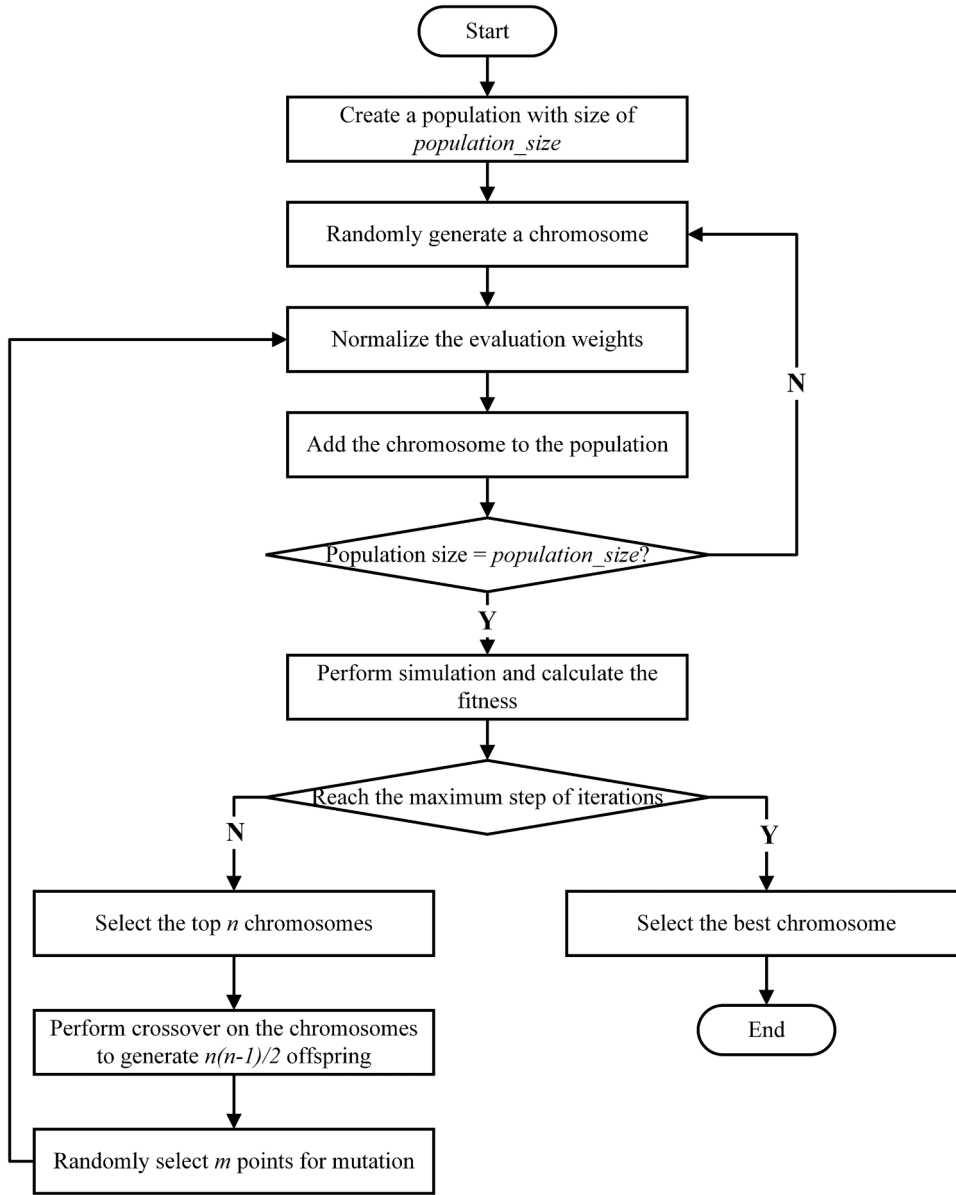


Fig. 14. Evaluation metric optimization process.

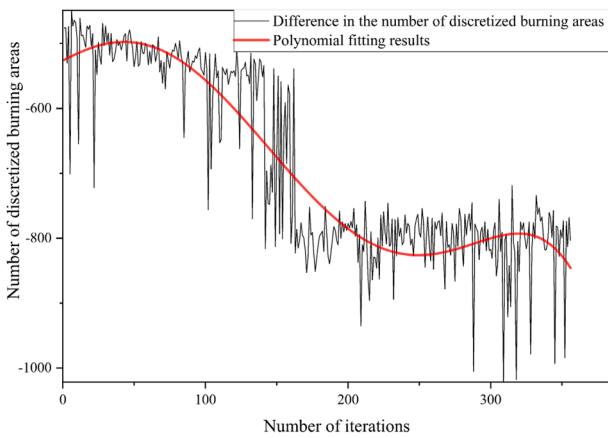


Fig. 15. Optimization results for water release strategy evaluation metric weights.

The multi-aircraft collaborative water release scenario includes the single-aircraft evaluation metrics from Fig. 8 and the multi-aircraft evaluation metrics from Figs. 10 and 11, such as fire line complexity and burning intensity. These metrics are directly encoded to form the "chromosomes" of the genetic algorithm. The optimization iteration process is illustrated in Fig. 14.

During crossover and mutation operations, the proportion of weights inherited from the previous generation's chromosomes is normalized. For any weight k_i , the normalization is performed as follows:

$$k'_i = k_i \cdot \sum_{k_m \in K_I} k_m / \sum k, k_i \in K_I \quad (25)$$

The natural fire spread result is used as a benchmark. The fitness of a chromosome is evaluated based on the total number of discretized burning and burned areas after the same duration of fire spread. Taking an example of 9 helicopters in 3 formations, the evaluation metrics for their water release strategies are optimized, yielding the results shown in Fig. 15. The optimization demonstrates that the firefighting effectiveness of multiple aircraft can be significantly improved.

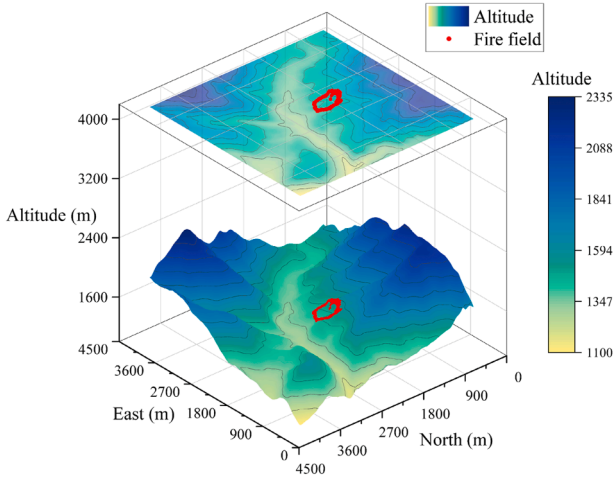


Fig. 16. Burning area for mission decision-making simulation.

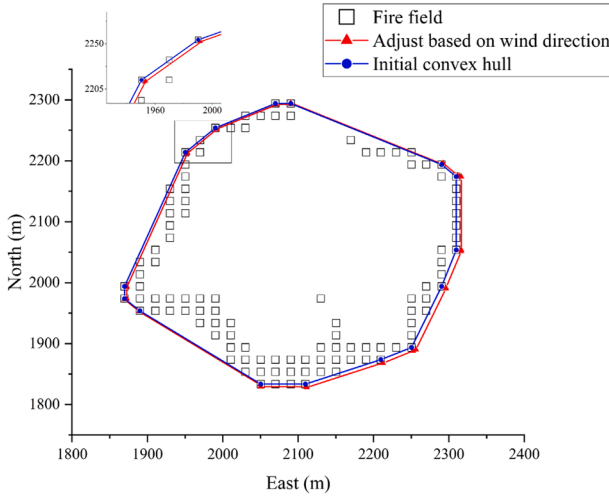


Fig. 17. Calculated envelope region for water release tasks.

4. Simulation and result

This section evaluates the proposed hierarchical decision-making model through three phases of testing. First, intra-formation decision behaviors are analyzed to demonstrate the computational process of decision-making. Second, multi-formation collaborative decisions are tested, with water release locations compared to spread-driven location selection strategies. Finally, mission effectiveness is assessed for different formation structures.

The simulation framework is implemented in Unity3D, where the physical process of fire spread is modeled using a cellular automata approach. Geographic Information System (GIS) data is integrated to derive terrain features, vegetation distribution, and water source locations within the operational area.

4.1. Simulation of multi-aircraft formation mission decision-making

The simulation scenario is based on the burning area shown in Fig. 16, with the fire-affected region spanning an altitude range of 1400.66 m to 1582.55 m and a total area of 42,800 square meters. The wind direction is set to 60° northwest.

The fire-affected area is enveloped, and the envelope is adjusted based on wind direction to determine feasible flight paths for water release. The results are shown in Fig. 17. The envelope is discretized into

equally spaced points, and each point is evaluated using the metrics outlined in Fig. 8. Since the simulation involves a single fire-affected area, the fire area metric is consistent across all points. In the simulation, blue line segments represent the envelope region (potential water release zones) derived from computational analysis of the discretized fire scene distribution. After accounting for wind direction effects on water release trajectories, the envelope region undergoes offset adjustments, resulting in red line segments denoting the aircraft flight paths. The effectiveness of water release locations formed by these red segments is thus analyzed.

A prior water release is assumed to have occurred at the coordinates (2102.03, 1833.59), with the aircraft heading east, a water dispersion width of 10 m, and a length of 50 m. The simulation results for each evaluation metric are shown in Fig. 18, with all metrics normalized for comparison. As illustrated in the figure, the metrics for wind direction alignment, water release zone overlap, and ground target angular deviation exhibit distinct clustered distributions. In contrast, other metrics demonstrate weaker regularity, which is attributed to irregular terrain undulations within the operational area.

Based on the single-aircraft evaluation, the decision-making results for concentrated water release and dispersed water release strategies are shown in Figs. 19 and 20, respectively. For concentrated water release, the maximum allowable overlap with previously released areas is set to 50 %. The formation enters the fire scene from the southwest direction and exits northeastward. As shown in the figure, under the dispersed water release strategy, individual aircraft execute water drops with significant spatial intervals to suppress multiple fire fronts across the scene. Conversely, under the concentrated water release strategy, aircraft perform sequential and overlapping water releases to intensively suppress the fire-affected area depicted in the figure.

Water release locations are selected based on overlap rate and, when the overlap rate is zero, the distance range from the previous water release location. For helicopters with limited water-carrying capacity or in scenarios involving intense fire fronts, an increased overlap rate may be prioritized to deliver concentrated water volume to localized areas. The dispersed water release strategy adapts according to the spatial density of the formation: tighter formations reduce the allowable distance range to minimize the need for aggressive maneuvers by wingmen.

4.2. Simulation of multi-aircraft and multi-formation mission decision-making

The multi-aircraft and multi-formation mission decision-making simulation is based on the two-layer hierarchical strategy framework proposed in this chapter. It aims to achieve coordinated decision-making at both the formation level and the individual aircraft level, simulating system-level tasks for forest firefighting missions.

The simulation involves multiple formations of different aircraft types, including fixed-wing water-dropping aircraft formations, bucket-equipped helicopter formations, and helicopters performing landing tasks. The experimental setup for the simulation is detailed in Table 1.

Based on the simulation setup, the water release positions and directions are determined as shown in Table 2. The azimuth angle is measured with 0° as true north and 90° as true east.

The relative positions of the water release points of each aircraft within the formation, the fire-affected area, and the landing point are illustrated in Fig. 21. The fire scene is depicted based on the state of the fire at the time of the last aircraft's water release in each formation. Additionally, the positions with the fastest fire spread in the fire spread model at the time of each aircraft's water release are marked as references for water release locations.

From Table 2 and Fig. 21, the following observations can be made:

- (1) Formation 1 executed a dispersed water release strategy, targeting smaller burning areas to control the development of the fire.

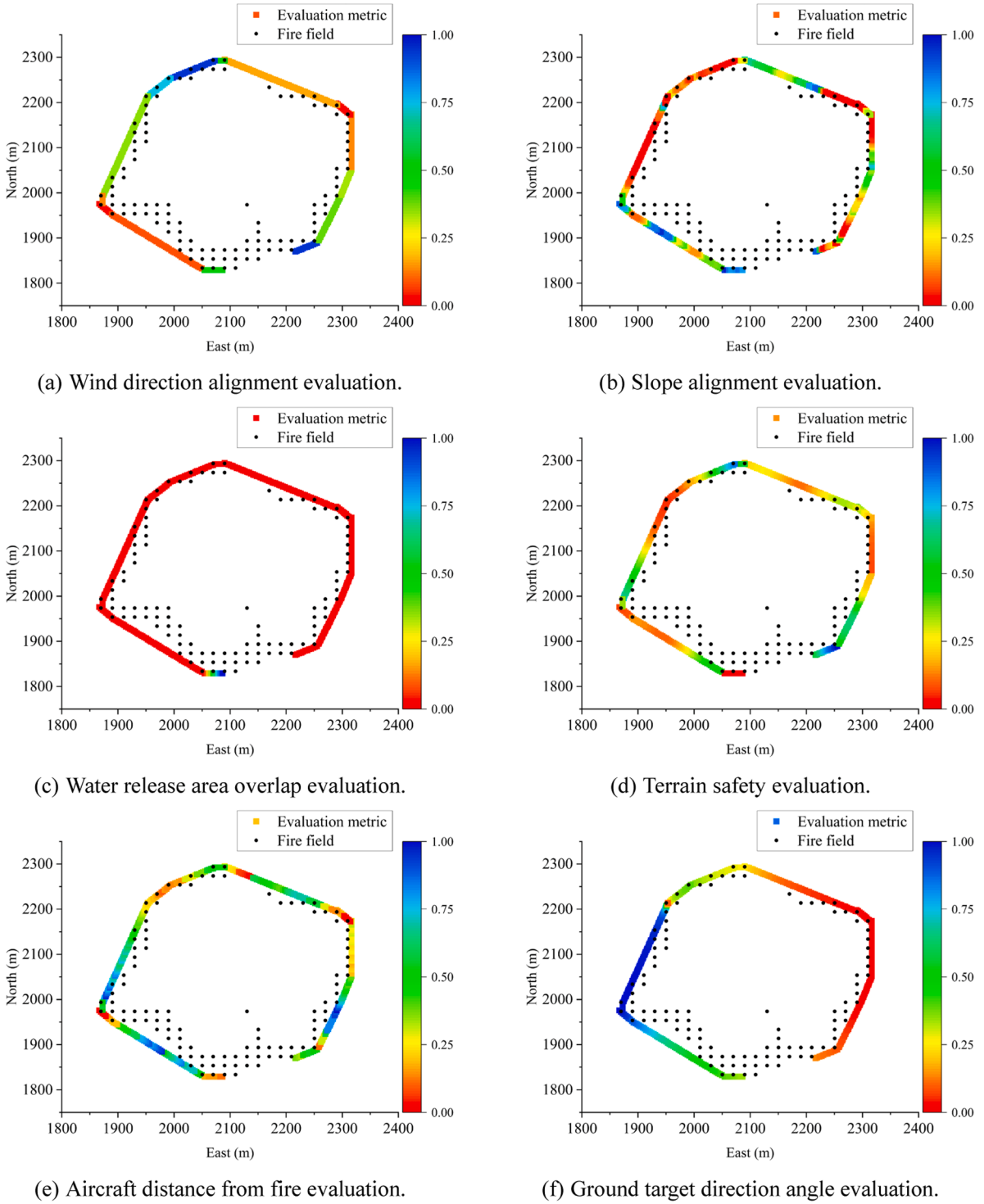


Fig. 18. Individual metric evaluation results.

- (2) Formation 2 implemented a follow-up overlapping water release strategy, concentrating water release at the tail of the fire to suppress its spread.
- (3) Formation 3 considered the relative position of the ground target and the fire-affected area, and since the distance between the ground target and the fire reached the predefined safety threshold (set as 500 m), Formation 3 executed an air-ground collaborative water release strategy to support ground personnel.

In the meanwhile, compared to the water release positions derived

from the forest fire spread model, the water release positions based on the multi-aircraft water release strategy model exhibit a spatial distribution that better aligns with the characteristics of formation-based water release. Additionally, the model can dynamically respond to the status of ground-protected areas, ensuring more effective and targeted fire suppression.

4.3. Simulation of aircraft utilization in forest firefighting missions

The mission scenario is simulated based on the area illustrated in

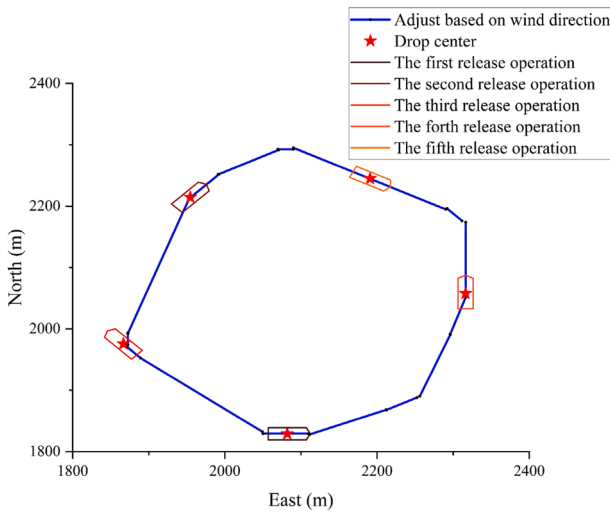


Fig. 19. Decision results for dispersed water release strategy.

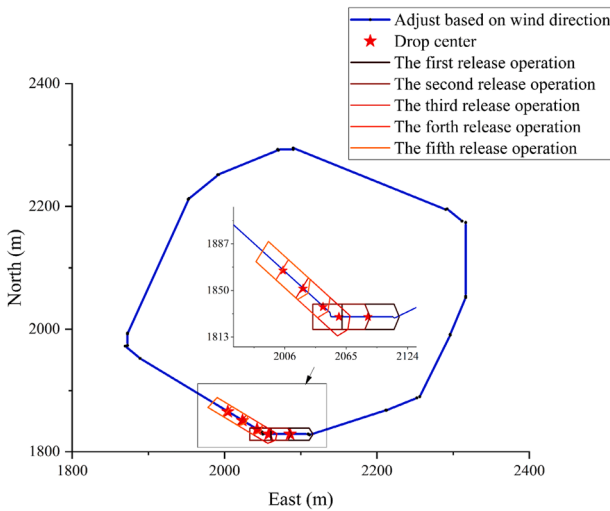


Fig. 20. Decision results for concentrated water release strategy.

Table 1
Simulation settings.

Parameter	Value
Number of formations	3
Formation 1 aircraft	4
Formation 1 aircraft type	Helicopter
Formation 2 aircraft	5
Formation 2 aircraft type	Fixed-Wing aircraft
Formation 3 aircraft	4
Formation 3 aircraft type	Helicopter
Number of ground target	1

Fig. 22, which includes three main regions: burning areas, water sources, and airports. The burning areas represent potential ignition points, with initial conditions set using the Monte Carlo method to simulate fires at different locations. The water sources and airports are selected from regions within the simulation area that meet the requirements for water collection and takeoff/landing, serving as boundary conditions for calculating aircraft entry/exit directions, flight range, and other parameters. These constraints influence the feasible water release directions and the number of water releases between two ground support operations.

Table 2
Multi-aircraft and multi-formation mission decision-making results.

Water Release Sortie	East Direction (m)	North Direction (m)	azimuth angle (°)
1-1	894.927	1123.083	116.5694
1-2	546.513	1123.362	-95.4413
1-3	369.548	659.911	135.0042
1-4	377.424	1107.259	84.55867
2-1	945.999	1146.33	-90
2-2	952.158	1141.513	-33.6852
2-3	978.616	1106.708	-36.8649
2-4	995.407	1079.696	-44.9948
2-5	986.323	1067.496	-86.9054
3-1	2950.225	3646.524	-94.4005
3-2	3061.939	3666.33	-90
3-3	3157.979	3666.33	-90
3-4	3255.689	3652.811	-78.6856

The specific coordinate information for the simulation scenario is provided in Table 3. The simulation scenario is set in mountainous terrain, necessitating terrain safety considerations for aircraft entering and exiting water release zones. The potential combustion area is situated between two water sources that meet refill requirements, allowing aircraft to select optimal water refill points post-operation based on flight distance and time constraints. Airport placement is determined using GIS data to identify larger nearby towns. The selection of simulation scenarios incorporates a degree of randomness, with variations in ignition locations across simulation rounds amplifying this stochasticity.

This section focuses on selecting helicopters for water release tasks, using Type A and Type B helicopters as examples. Both are medium-sized helicopters capable of performing bucket-based firefighting missions. Their performance comparison is provided in Table 4.

The task effectiveness of Type B helicopters is analyzed and evaluated by constructing a multi-formation setup with three formations, each consisting of 3 helicopters. Based on the inter-formation collaboration strategies and single-aircraft strategies developed in the previous chapter, the following operational modes are adopted: Formation 1 executes a dispersed water release strategy to control the fire. Formation 2 and Formation 3 follow up with concentrated water release on fire-affected areas.

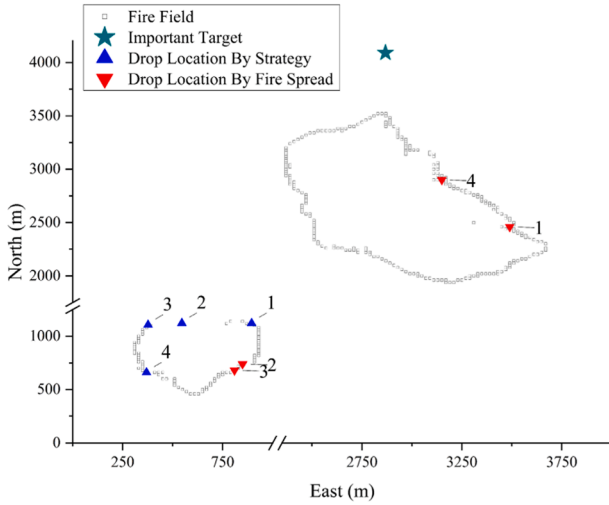
In terms of aircraft configuration, Formation 0 uses 9 Type A helicopters as a reference, while the other formations each consist of 1 Type B helicopter and 8 Type A helicopters. The Type B helicopter assumes different roles in the formations, serving as the lead aircraft, wingman and second wingman in different formations. The total duration of a single simulation round is set to 6 hours, and each aircraft performs at least one return-to-base support operation during the simulation. The changes in the number of discretized burning regions during the single-round simulation are shown in Fig. 23, with the position of the Type B helicopter in the simulation distinguished using the format "Formation Number-Position Number."

The changes in the number of discretized burning regions were recorded, and the following fire-related metrics were evaluated at the end of each simulation:

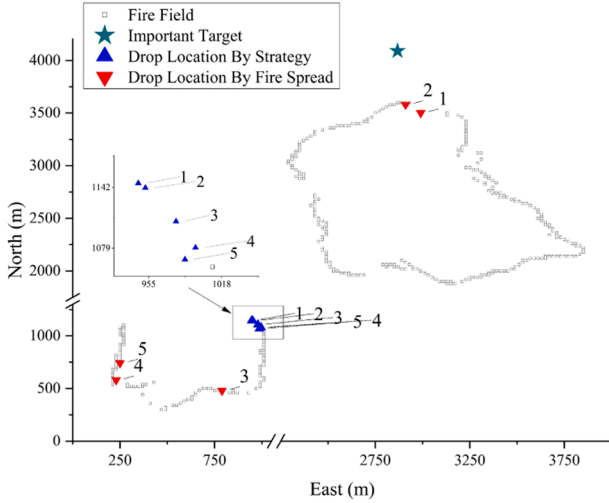
- (1) Burning area: The area still actively burning.
- (2) Burned area: The area that has been burned.
- (3) Total fire area: The sum of the burning and burned areas.
- (4) Water release coverage: The area effectively covered by water release.

In this evaluation framework, burning area, burned area, and total fire area are considered optimal when minimized, while water release coverage is considered optimal when maximized. These criteria are used to assess the relative firefighting effectiveness of different formation structures in each simulation round.

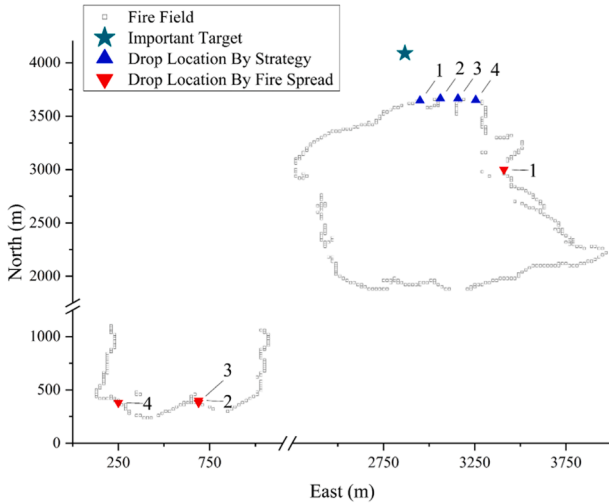
The impact of water release altitude on simulation outcomes is



(1) Formation 1 water release results.



(2) Formation 2 water release results.



(3) Formation 3 water release results.

Fig. 21. Multi-aircraft and multi-formation mission decision-making results.

analyzed through a series of tests. Using a 50-meter release altitude as the baseline for water discharge zone width, simulations are conducted for altitudes ranging from 30 to 70 m, yielding the analytical results presented in Tables 5–8.

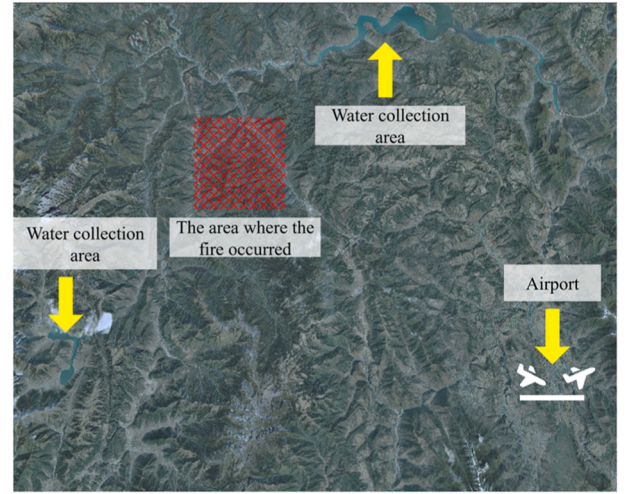


Fig. 22. Simulation scenario area.

Table 3
Simulation Scenario Area Information.

Attribute	Value	Unit
Burning area		
Minimum longitude	102.3125	°
Maximum longitude	102.4375	°
Minimum latitude	29.06279	°
Maximum latitude	29.18824	°
Longitudinal length	12.1419	km
Latitudinal length	13.9508	km
Water source 1		
Longitude	102.2041	°
Latitude	28.90636	°
Water collection azimuth	−171.42	°
Water source 2		
Longitude	102.6163	°
Latitude	29.32848	°
Water collection azimuth	124.33	°
Airport		
Longitude	102.7421	°
Latitude	28.843	°
Runway orientation	−20	°
Simultaneous aircraft support	4	-

Table 4
Parameters for two types of helicopters.

Parameter	Type A	Type B
Maximum Takeoff Weight(kg)	13,000	11,000
Bucket Water Capacity(kg)	2500	4000
Maximum Range(km)	570	800
Maximum Cruise Speed(km/h)	250	260
Cruise Speed(km/h)	230	230
Maximum Endurance Time(h)	2.5	3

The standard deviation values indicate that water release altitude affects all result metrics, but for different formation structures, these deviations remain within the same order of magnitude. This suggests that the influence of altitude on formation firefighting effectiveness is consistent across formation types. The trend slopes reflect the directional changes in result metrics with varying altitudes. Overall, as altitude increases, all metrics exhibit a decreasing trend. This is attributed to two factors: (1) higher altitudes expand the coverage area of the water discharge zone, and (2) reduced water volume per unit area necessitates increased reliance on overlapping water release tactics.

Furthermore, sensitivity analysis is performed by calculating the coefficient of determination R^2 and t -test result P for regression

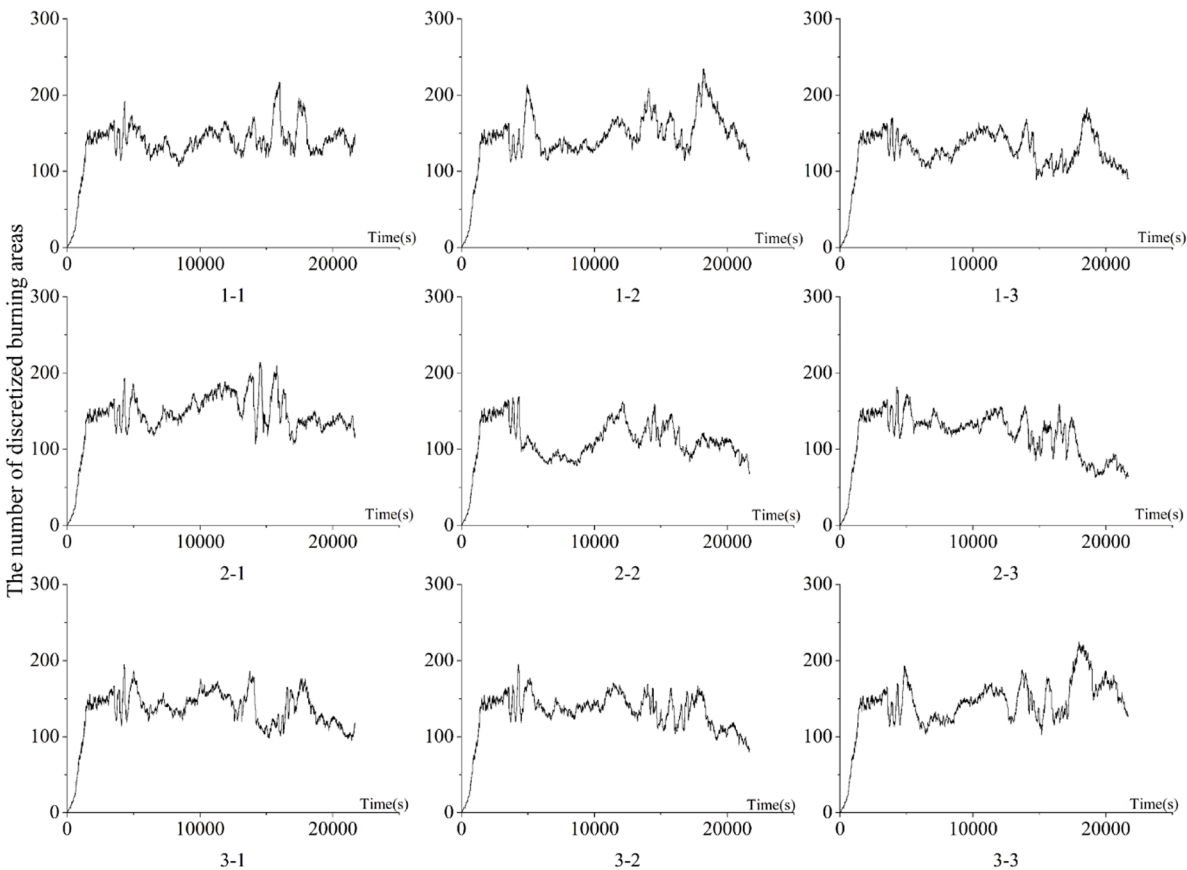


Fig. 23. Changes in the number of discretized burning regions over time in a single simulation.

Table 5
Sensitivity analysis results of burning area quantity under varying water release altitudes.

Fleet	Order	Final rank	Standard deviation	Value trend	<i>P</i>	<i>R</i> ²
1	1	1	7.11765	−0.47051	0.64112	0.03277
1	2	3	7.29391	0.90641	0.37017	0.11582
1	3	2	11.92281	−0.45342	0.78974	0.01085
2	1	4	11.34861	−2.10897	0.16174	0.25901
2	2	6	6.97369	−0.97949	0.30669	0.14796
2	3	5	9.90558	−1.53846	0.25374	0.18092
3	1	8	5.97248	−1.36667	0.07092	0.39271
3	2	9	8.37374	−0.69829	0.55451	0.05215
3	3	7	8.90535	−0.02821	0.98233	7.5234E-05

Table 6
Sensitivity analysis results of burned area quantity under varying water release altitudes.

Fleet	Order	Final rank	Standard deviation	Value trend	<i>P</i>	<i>R</i> ²
1	1	1	99.01482	−8.46325	0.54437	0.05479
1	2	3	90.69092	−1.56368	0.90399	0.00223
1	3	2	116.92863	−3.89829	0.81529	0.00834
2	1	4	95.83457	−5.18333	0.70371	0.02194
2	2	5	93.58733	2.34829	0.86056	0.00472
2	3	7	102.34025	−10.23120	0.47592	0.07496
3	1	9	99.84974	−20.90256	0.10657	0.32867
3	2	8	116.26296	11.02179	0.49992	0.06740
3	3	6	96.99616	4.15684	0.76363	0.01377

Table 7
Sensitivity analysis results of total fire area quantity under varying water release altitudes.

Fleet	Order	Final rank	Standard deviation	Value trend	<i>P</i>	<i>R</i> ²
1	1	1	102.90648	−8.93376	0.53790	0.05653
1	2	3	94.35831	−0.65726	0.96115	0.00036
1	3	2	126.78971	−4.35171	0.80992	0.00884
2	1	4	104.43731	−7.29231	0.62212	0.03657
2	2	5	94.97897	1.36880	0.91970	0.00156
2	3	7	110.63219	−11.76966	0.44687	0.08488
3	1	9	103.03655	−22.26923	0.09314	0.35034
3	2	8	120.24048	10.32350	0.54252	0.05529
3	3	6	103.43697	4.12863	0.77952	0.01195

Table 8
Sensitivity analysis results of water release coverage quantity under varying water release altitudes.

Fleet	Order	Final rank	Standard deviation	Value trend	<i>P</i>	<i>R</i> ²
1	1	1	15.68186	−1.50299	0.49504	0.06889
1	2	2	19.69898	−2.52821	0.35365	0.12354
1	3	3	17.85220	−2.70855	0.26606	0.17264
2	1	4	14.62568	−1.15641	0.57575	0.04689
2	2	5	12.23217	−1.88590	0.25760	0.17827
2	3	6	9.95693	−1.01410	0.46734	0.07780
3	1	7	9.83952	−0.56581	0.68574	0.02480
3	2	8	8.24072	−0.02521	0.98293	7.0211E-05
3	3	9	11.16999	−1.41795	0.35929	0.12086

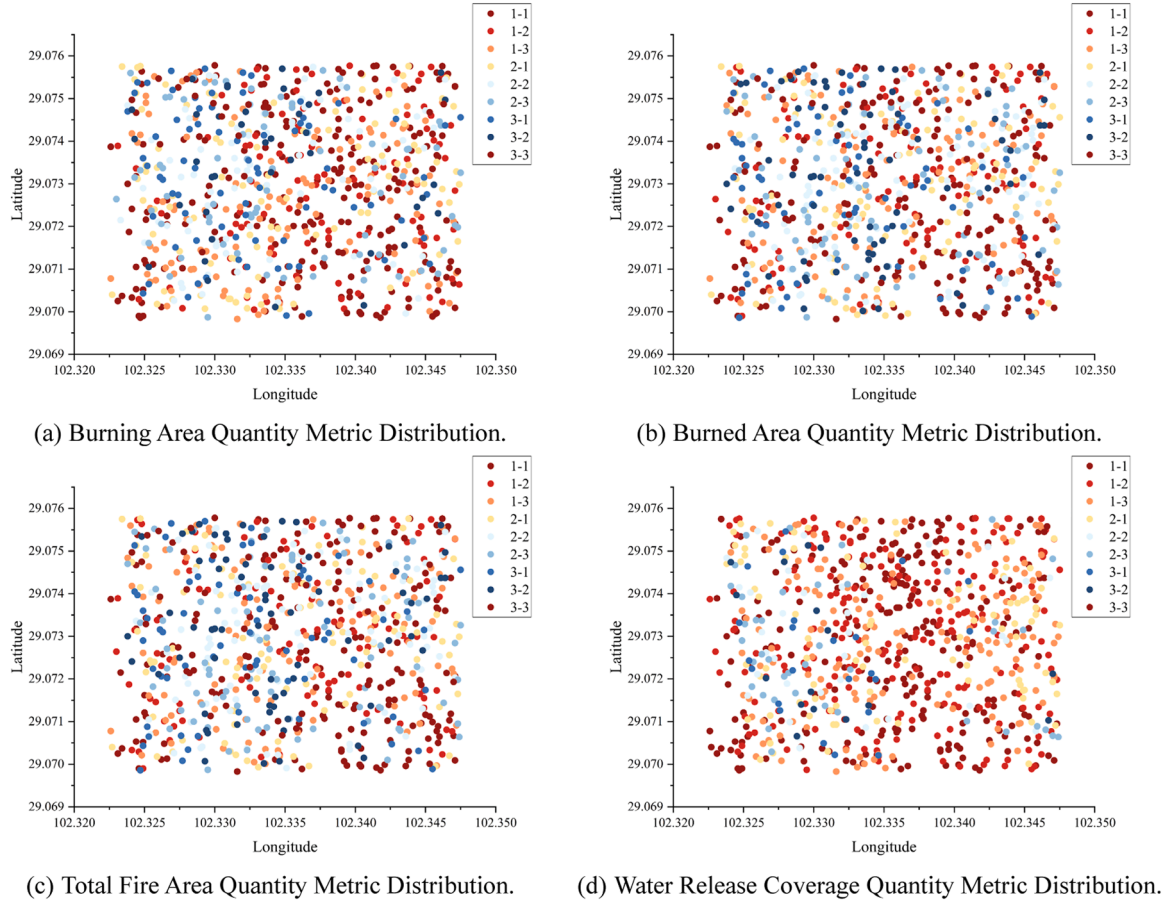


Fig. 24. Spatial distribution of results for different metrics.

Table 9

Calculation results of geospatial sensitivity for simulation outcomes.

Fleet type	Burning area result		Burned area result		Total fire area result		Water release coverage result	
	With longitude	With latitude	With longitude	With latitude	With longitude	With latitude	With longitude	With latitude
1-1	0.12134	0.00310	0.16944	-0.01526	0.17041	-0.01457	0.03394	-0.03690
1-2	0.09984	-0.06735	0.15548	-0.11626	0.15584	-0.11616	0.15860	0.10668
1-3	0.16034	-0.03285	0.15762	-0.11456	0.16057	-0.11238	0.17579	0.10793
2-1	0.14293	-0.08671	0.18464	-0.10291	0.18578	-0.10392	0.20363	0.19410
2-2	0.19933	-0.10948	0.18263	-0.10436	0.18673	-0.10648	0.14519	0.16243
2-3	0.14174	-0.06583	0.21300	-0.09841	0.21324	-0.09854	0.16949	0.21452
3-1	0.08710	-0.11043	0.22337	-0.10505	0.22029	-0.10726	0.24549	0.27698
3-2	-0.00267	-0.19413	0.18016	-0.13689	0.17379	-0.14301	0.30707	0.30269
3-3	-0.08382	-0.24468	0.14616	-0.16260	0.13627	-0.17066	0.34695	0.40246

coefficients. The results demonstrate strong sensitivity of simulation outcomes to water release altitude. However, the R^2 values indicate that altitude variations only partially explain the differences in outcomes across simulation groups. This partial explanatory power arises from geographical disparities between simulation groups, which inherently affect fire spread dynamics and firefighting efficacy.

In summary, while water release altitude influences simulation results, its impact is consistent across different formation structures. Consequently, altitude settings do not affect comparative assessments of formation firefighting effectiveness.

Analysis of multi-round simulation results based on initial ignition locations yields the distribution illustrated in Fig. 24. The results show no geospatial specificity in the optimal formation structures across evaluation metrics, indicating that simulation outcomes are not clustered by geographical location. The Pearson correlation coefficient (PCC) is applied to assess the correlation between simulation results and

the latitude/longitude of initial ignition points, as shown in Table 9. While the ignition location influences the firefighting effectiveness of all formations, correlation coefficients remain below 0.5, suggesting a weak spatial dependency. Consequently, the simulation results demonstrate a degree of universality, with firefighting effectiveness being minimally affected by geographical variations.

Then, a numerical analysis of the results is conducted. First, the frequency of achieving optimal metrics for each formation structure across simulation rounds is statistically summarized, as shown in Fig. 25. Based on Fig. 25, Formation 1-1 demonstrates significant superiority across all four metrics, indicating its cross-metric stability. From Formation 1 to Formation 3, the optimal frequency for each metric exhibits an exponential decay trend, with the steepest decline slope observed for the water release coverage metric, highlighting its heightened sensitivity to formation structures. Within formations, systematic reductions in optimal frequency are observed as the sequence of

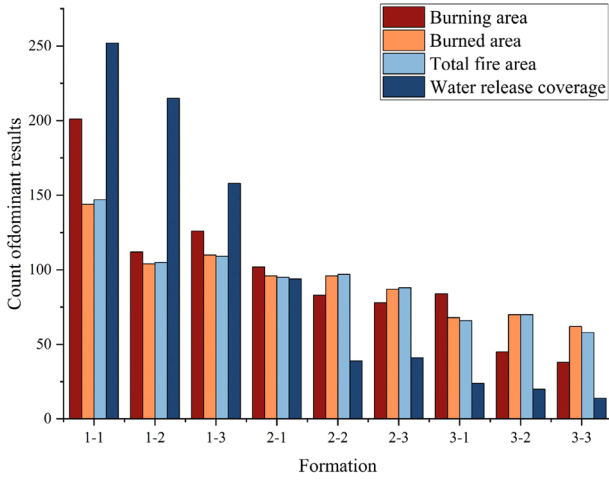


Fig. 25. Distribution of optimal formation configurations in multi-round simulations.

Type B helicopters increases. Additionally, a strong correlation is identified between the water release coverage metric and the other three metrics (Pearson correlation coefficient $r = 0.89$, significance level $p < 0.01$), which will be further discussed in subsequent analyses.

Using a baseline formation of 9 Type A helicopters as a reference, the simulation results for the 9 formation configurations with 1 Type B helicopter were compared. The differences between these results and the baseline are shown in Fig. 26.

Based on the above results, Formation 1-x demonstrates significant advantages across all metrics, with the 1-1 formation consistently

occupying the optimal position. This indicates that such a formation inherently coordinates well in suppressing fire spread and reducing burned areas, while also excelling in water release coverage, as shown in Fig. 26(d). Overall, the quantity of water release coverage exhibits a strong correlation with fire spread containment effectiveness ($r = 0.92$, $p < 0.01$), aligning with the earlier analysis of outcome dominance frequency. This correlation may stem from tactical preferences for sequential overlapping releases or dispersed water drops in decision-making, which suppress fire spread across broader areas. The observed correlation also provides direction for further optimization of firefighting strategies.

From the simulation results, when Type B helicopters occupy the lead aircraft or first wingman positions in Formation 1, optimal outcomes are achieved in both fire suppression and water coverage. In practical firefighting operations, commanders often prioritize deploying helicopters with larger water-carrying capacities for initial fire suppression. The simulation experiments conducted in this section provide numerical validation for this empirical practice.

5. Conclusions

The participation of multiple aircraft in forest firefighting operations enables the delivery of a larger volume of fire retardants to the fire-affected area in a shorter time compared to single-aircraft operations. However, coordination between aircraft and formations is a critical yet often overlooked aspect of multi-aircraft firefighting research.

This paper proposes a multi-aircraft formation water release decision-making model that addresses water release constraints and constructs a two-layer decision-making framework, ranging from single-aircraft strategies to formation-level strategies. The model incorporates a quadtree-based two-dimensional data indexing method for efficient

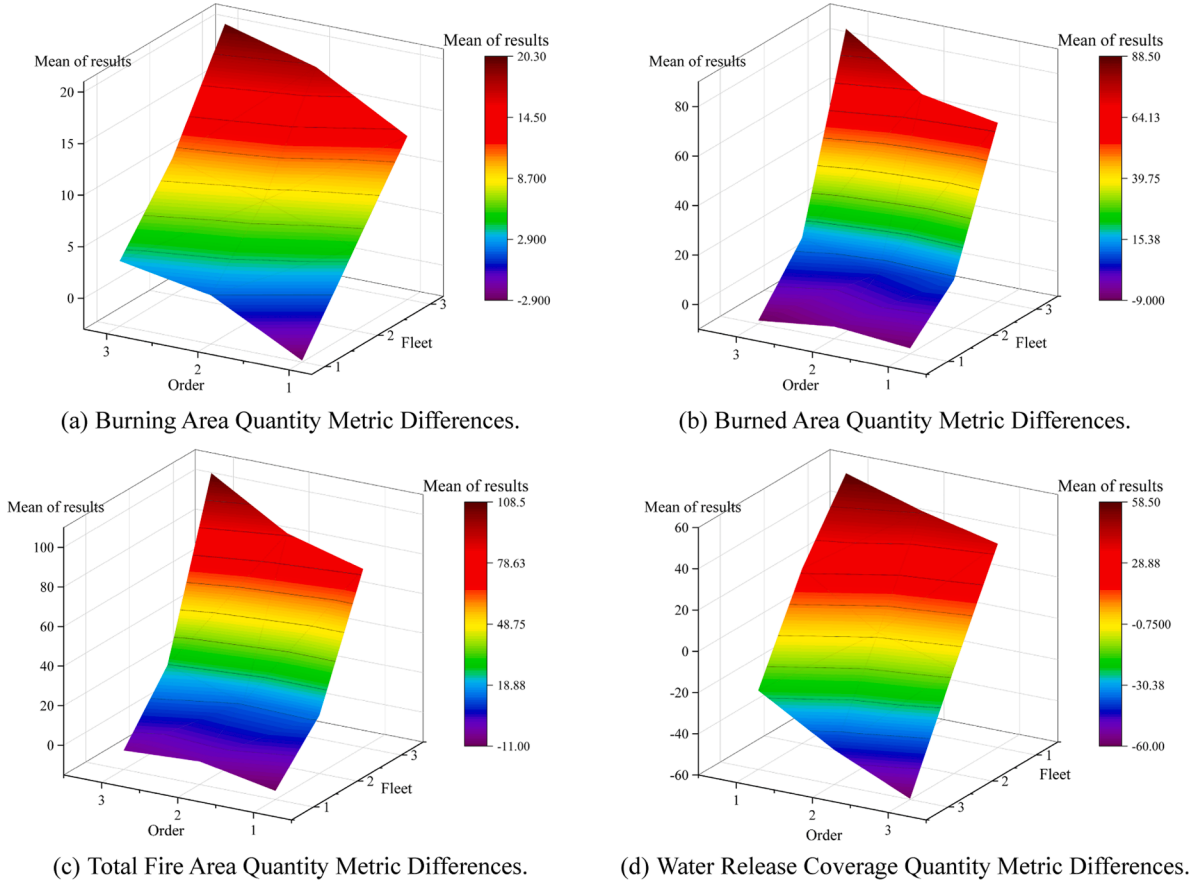


Fig. 26. Differences in result values.

fire scene data processing and utilizes a genetic algorithm to optimize decision metrics. The simulations conducted in Section 4 demonstrate quantitative outcomes of the decision-making process, highlighting the tactical coordination achieved through the proposed method. Comparative experiments on formation structures further analyze the sensitivity of water release altitudes and fire scene locations, yielding evaluation results that align with empirical firefighting practices.

Future work will focus on:

1. Deepening exploration of the metric weight parameter space: Implementing experimental design methods to sample metric weights and construct a comprehensive mapping dataset between weights and firefighting effectiveness. Global optimization techniques, such as swarm intelligence algorithms and gradient descent, will be applied to identify optimal weight configurations.
2. Human-in-the-loop simulations with firefighting commanders: Collaborating with aerial firefighting command personnel to conduct human-in-the-loop simulations, followed by comparative analysis of discrepancies between model-driven and human decisions.
3. Integration with detailed aircraft modeling: Combining the decision model with high-fidelity aircraft performance models to investigate the impact of design parameters (e.g., payload capacity, maneuverability) on multi-aircraft formation firefighting effectiveness. Sensitivity analyses will be performed to evaluate how variations in aircraft specifications influence mission outcomes.

CRedit authorship contribution statement

Hu Liu: Supervision, Methodology, Conceptualization. **Siliang Liu:** Writing – original draft, Software, Methodology, Conceptualization. **Yongliang Tian:** Writing – review & editing, Validation, Conceptualization. **Kangwen Sun:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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